

ChE422 Biochemical Engineering

2020 Midterm Exam

3/10/2020

2:30 PM - 3:45 PM

Name:



Instructions: This is a close-book and close-note exam.

MULTIPLE CHOICE QUESTIONS (Part I):

Note: Only ONE answer is correct. Circle the correct answer. (2 pts for each problem)

- 93/100
- The bond between a phosphate group and the ribose sugar group in RNA is called which of the following?
(A) Phosphodiester linkage ✓
(B) Glycosidic bond
(C) Peptide bond
(D) Amide bond ✓
 - Which of the following plays a role in DNA transcription?
(A) DNA polymerase
(B) RNA polymerase ✓
(C) DNAase I
(D) Guanyl transferase ✓
 - What signals the end of transcription?
(A) Stop codon (translation)
(B) Terminator ✓
(C) The end of the DNA chain ✓
(D) RNA polymerase runs out
 - The start codon is:
(A) AUU (B) UAA (C) AUG ✓ (D) UAG ✓
 - Which of the following is not a component of a nucleotide?
(A) Phosphate group
(B) Anti-codon ✓
(C) Ribose sugar
(D) Nitrogen base
 - What provides the energy that drives the addition of nucleotides to a growing DNA chain during replication?
(A) The hydrolysis of peptides
(B) The hydrolysis of the diphosphate group in dNTPs ✓
(C) The hydrolysis of UTP
(D) The hydrolysis of GTP ✓

7. Which of the following is the sequence of the -10 region of prokaryotic promoters?

- (A) TATAAT (B) TTGACA (C) TATA (D) CAA

✓

8. How many base pairs make up a codon?

- (A) 1 (B) 2 (C) 3 (D) 4

✓

✓

9. How many amino acids are there?

- (A) 20 (B) 30 (C) 40 (D) 50

✓

✓

10. How many hydrogen bonds are formed between one A:T base pair?

- (A) 1 (B) 2 (C) 3 (D) 4

✓

✓

MULTIPLE SELECTION QUESTIONS (Part II):

Note: ONE or MORE answers can be correct. Circle all correct answers. (3 pt for each problem)

11. Which of the following are the protein secondary structures?

- (A) Hairpin (B) α -helix (C) β -Sheet (D) Peptide bond

✓

12. Which of the following in a plasmid controls the copy number of this plasmid in the host?

- (A) Antibiotic resistance gene ✗
(B) Multiple cloning sites ✗
(C) LacZ gene ✗
(D) Origin of replication (Ori)

✓

✓

13. Which of the followings are required for Polymerase chain reaction (PCR)?

- (A) DNA template (B) T4 ligase (C) DNA polymerase (Taq) (D) DNA primers

✓

14. Which of the following methods are commonly used to transform plasmid into *E. coli* cells?

- (A) Cell fusion
(B) Heat shock method
(C) Electroporation
(D) DNA injection by a needle

✓

✓

15. Which of the followings are the key components in a general Operon?

- (A) Promoter (B) Operator (C) phage gene (D) Structural genes

✓

16. Which of the following methods are used in random mutagenesis?

- (A) Error prone PCR
- (B) Chemical Deamination
- (C) DNA shuffling
- (D) *E. coli* XL1 Red method

17. Agarose gel electrophoresis is able to separate linear DNA fragments with which of the following parameter?

- (A) Charge
- (B) Size
- (C) Secondary structure
- (D) Conformation

18. In order to use pET expression system for protein expression, the host cell should provide with which of the following component for transcription?

- (A) Taq DNA polymerase
- (B) T7 polymerase
- (C) protease
- (D) Histag

19. Which of the following methods can be used to introduce the mutation at the specific sites in a gene?

- (A) UV irradiation
- (B) Cassette mutagenesis
- (C) PCR site-directed mutagenesis
- (D) RACHITT method

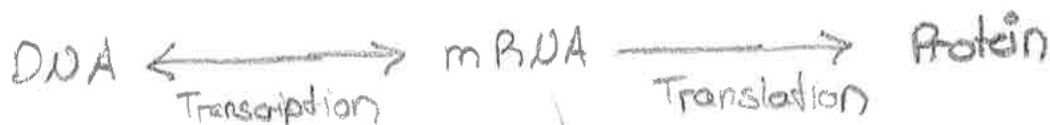
20. Which of the following compound is used in pBAD expression system to induce protein expression?

- (A) Glucose
- (B) IPTG
- (C) L-arabinose
- (D) Lactose

Brief answer the following questions

1. What is Central Dogma of Molecular Biology? (5pts)

The central dogma of molecular biology describes the flow of information from DNA to the final protein produced, see figure below:



2. List three key differences between prokaryote and eukaryote? (5pts)

① Prokaryotes have circular DNA, eukaryotes have linear DNA. ✓

② Eukaryotes can perform post translational modification, Prokaryotes cannot perform this function.

③ Prokaryotic cells you can insert plasmids that can independently replicate inside cells, eukaryotic cells must have oriA sequence directly inserted into gene.

3. Describe the general key components of an expression plasmid? (5pts) (See Note on Last Page!)

Ori - site of origin, this site controls plasmid replication and copy number ✓

Antibiotics Resistance Genes - genes that produce antibiotic resistance molecules for the bacteria.

MCS - multiple cloning site, a site where you can easily insert your desired genes into the plasmid.

4. Describe the key differences between blunt-end ligation and sticky end ligation for cloning? (5pts)

In blunt-end ligation, since there are no overhangs, there is no affinity for the ends to attach in any particular way. ✓

Therefore, the efficiency is low and direction of insertion cannot be controlled. In sticky end ligation, there are overhangs which increase the affinity of the DNA end to attach to the corresponding base pair of another DNA strand. Sticky end ligation is much more efficient and direction of insertion can be controlled.

5. List at least two methods to confirm the insertion of a gene into the expression plasmid? (5pts)

① SDS-PAGE - You will see a large thick band on gel, signifying a large expression of protein, plus you can check molecular weight against your protein's known weight.

② Western Blot Test - You use an antibody conjugated to an enzyme that specifically targets your protein. Add substrate and if it reacts then you have inserted your protein. ✓

6. Briefly describe blue-white screen? (10pts)

The blue-white screen is a method to confirm that the plasmid with inserted gene has been transformed into the bacteria. The bacteria that have the inserted gene will disrupt B-galactosidase gene and render it dysfunction. The bacteria without the inserted gene will have a functional B-galactosidase gene. The blue-white screen introduces a chemical that only a functional B-galactosidase can turn blue. Therefore the white colonies will most likely have your inserted gene. Bacteria that don't have the transformed plasmid will not grow as they won't have antibiotic resistance genes.

7. After you find an interesting gene in a new organism, briefly describe the procedures to let *E. coli* express this protein. (15pts)

① Sequence Protein

- Use mass spectroscopy or Edman degradation to determine primary of Amino Acid

② DNA sequence

- Reverse translate Amino Acid sequence into DNA sequence
- Remove all known introns from sequence

③ DNA into Plasmid

- Use a restriction enzyme to create sticky ends and linearize the plasmid. Use pET28a(+) as protein expression plasmid.
- Insert your desired DNA strand with a DNA ligase to reconnect DNA and make the plasmid circular again.

④ Transform Plasmid into E. coli

- Use heat shock method to get plasmid into *E. coli* cells. Use the GM0 *E. coli* cells with DE3 genes for T7 RNA polymerase.

⑤ Let Bacteria Grow

- Let bacteria grow to a large enough size to produce proteins.

⑥ Introduce IPTG

- Introduce IPTG (lactose analog) to deactivate the LacZ inhibitor and allow T7 RNA polymerase to start transcribing your genes and then produce your proteins.

⑦ Extract Protein

- Use sonication to lyse the *E. coli* cells and centrifuge to extract the cell debris.

⑧ Protein Confirmation

- Use SDS-PAGE to get the molecular weight of protein and compare to the known protein weight.

⑨ Isolate Protein

- Use IMAC or chromatography to isolate and purify the protein and check for functionality with its substrate. If functional, you have your desired protein.

★ See back!